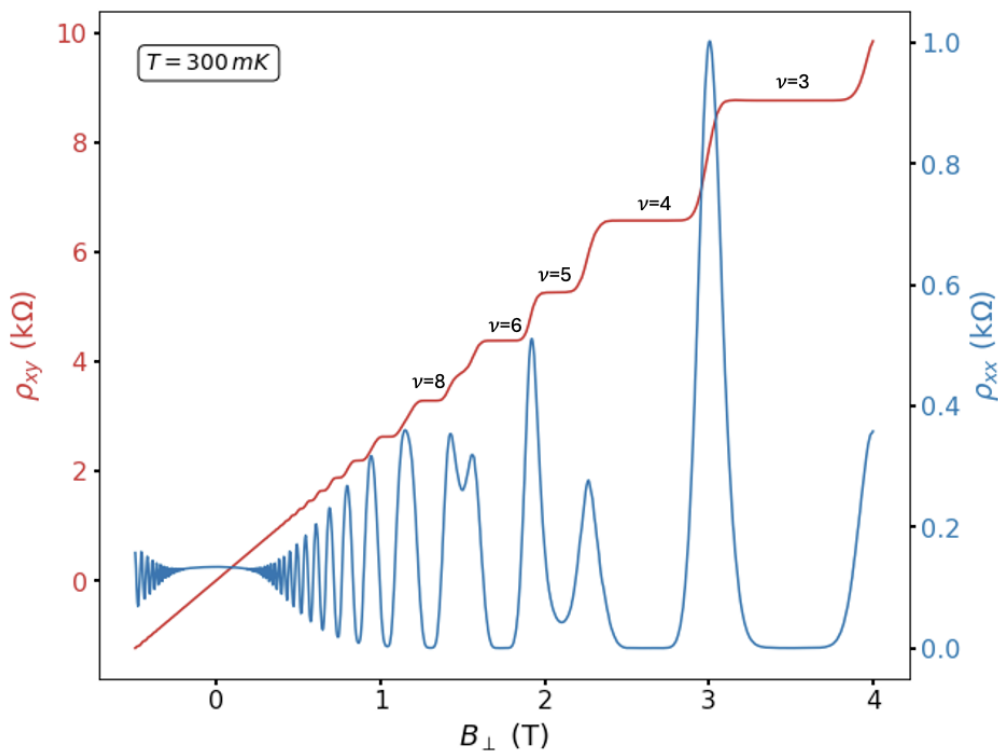




BSc Project in Physics

Teaching and Understanding the Quantum Hall Effect Experimentally

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Abstract

This project investigates how the newly established Quantum Training Lab (QTL) at the Niels Bohr Institute can be used to teach students about quantum phenomena through hands-on experimentation. The quantum phenomenon studied in this project was the quantum Hall effect (QHE), specifically the integer quantum Hall effect. An experiment exploring the QHE served as the final experiment for students enrolled in the course “Quantum Phenomena in Nano Systems (Nano3)”. A group of students from Nano3 were observed while they performed the QHE experiment and afterwards interviewed. Their responses and the observations were used to evaluate the educational value of the QTL, and the structure of the experiment.

The implementation of the QHE experiment and the use of the QTL within the Nano3 course worked well overall, with students demonstrating strong engagement and improved understanding of key quantum concepts. However, the current format is highly dependent on teaching assistants and includes significant downtime, making it impractical for larger courses without restructuring.

In addition to the students being observed and interviewed, the sample used by the students was characterized as part of the project. The most notable extracted value was the extracted effective mass of the electrons, found to be $m^* = (0.088 \pm 0.001) m_0$. This value suggests that the wavefunction of the electrons in the sample, a GaAs/AlGaAs heterostructure, is predominantly localized in the GaAs layer.

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1 Introduction

Quantum mechanics has been a cornerstone of modern physics for a century, and remains an integral part of any upper-level physics education today. As quantum technology is becoming more widespread and expanding beyond research laboratories into industry, a qualified and sufficiently large quantum workforce is crucial [1]. Therefore educators and academic institutions have the responsibility to attract and prepare more students to enter the quantum workforce.

Quantum mechanics instruction has commonly focused on the mathematical framework of the science [2]. Experiments that demonstrate quantum phenomena are not always accessible, since they often require expensive and advanced equipment.

In order to offer students the opportunity to learn about quantum mechanics experimentally, the Niels Bohr Institute recently established a new laboratory called the Quantum Training Lab (QTL). By providing students access to the same state-of-the-art infrastructure that is available in research laboratories, they will be able to perform more experiments involving quantum phenomena. By improving and modernizing the teaching framework for the students, it will hopefully inspire more talented young people to work and do research in quantum physics.

The QTL was put into use for the first time in the spring of 2025. Students enrolled in the course "Quantum Phenomena in Nano Systems (Nano3)" carried out several preliminary experiments before finishing the course with an experiment that explored the quantum Hall effect (QHE).

The purpose of this project is to investigate the use of the QTL. What do students think about it and how can it help promote the education of the next generation of quantum physicists? In order to do so, one of the groups from Nano3 was observed during the experiment. Afterwards, the students in the group were interviewed.

Before looking at the results from observing and interviewing the students, it is important to understand the theory behind the QHE. It is also necessary to understand the experimental setup and the tools used to carry out the experiment.

In chapters 2-4 the theory behind the quantum Hall effect will be explained. This is followed by chapter 5, which provides a detailed explanation of how to set up and carry out the experiment, as well as an explanation of how the various instruments used in the experiment work. Chapter 6 contains the results from observing and interviewing the Nano3 students. Chapter 7 contains the experimental results from when I performed the experiment myself. In chapter 8 the results from the student study and ideas for improving the experiment and the course are discussed. Chapter 9 concludes the project. Appendix A contains a list of variables used in the project.

To fully benefit from reading this thesis, the reader is expected to have a basic understanding of quantum mechanics and solid state physics.

2 The classical Hall effect

The classical Hall effect appears in conductors that have a current running through them while being subject to a magnetic field perpendicular to the current. The Lorentz force acts on the electrons and causes them to drift to one side of the conductor. One side of the conductor

becomes negatively charged, while the opposite side becomes positively charged. This causes a transverse voltage difference U_{xy} , which is also called the Hall voltage [3].

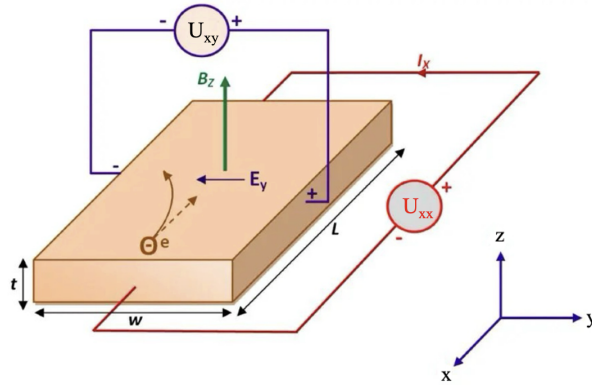


Figure 2.1: Classical Hall effect [4].

Mathematically, the Hall effect can be described using the Drude model, where electrons in a solid are considered free. The electrons do not experience Coulomb interactions and move freely for a time τ between collisions. When such an electron, with effective mass m^* and velocity v_d , is placed in an electromagnetic field, it obeys the following equation of motion

$$m^* \frac{d\mathbf{v}_d}{dt} = -e(\mathbf{E} + \mathbf{v}_d \times \mathbf{B}) - \frac{m^* \mathbf{v}_d}{\tau}. \quad (2.1)$$

In the stationary state there is no change in velocity. For a two-dimensional system the electric field is

$$\begin{pmatrix} E_x \\ E_y \end{pmatrix} = \begin{pmatrix} -m^*/e\tau & -B \\ B & -m^*/e\tau \end{pmatrix} \begin{pmatrix} v_x \\ v_y \end{pmatrix} \quad (2.2)$$

$$= \begin{pmatrix} m^*/e^2 n \tau & B/ne \\ -B/ne & m^*/e^2 n \tau \end{pmatrix} \begin{pmatrix} J_x \\ J_y \end{pmatrix} \quad (2.3)$$

where the current density $\mathbf{J} = -ne\mathbf{v}_d$, with n being the density of charge carriers, is used. (2.3) is actually Ohm's law $\mathbf{J} = \sigma\mathbf{E}$. It is written in the form

$$\mathbf{E} = \rho\mathbf{J}, \quad (2.4)$$

with ρ and σ being the resistivity and conductivity tensors. They satisfy the identity $\rho = \sigma^{-1}$, and in two dimensions they take the form [3]

$$\sigma = \begin{pmatrix} \sigma_{xx} & -\sigma_{xy} \\ \sigma_{xy} & \sigma_{xx} \end{pmatrix} \text{ and } \rho = \begin{pmatrix} \rho_{xx} & \rho_{xy} \\ -\rho_{xy} & \rho_{xx} \end{pmatrix}. \quad (2.5)$$

By tensor inversion expressions for the components of the resistivity tensor, ρ_{xx} and ρ_{xy} , are obtained [3]

$$\rho_{xx} = \frac{\sigma_{xx}}{\sigma_{xx}^2 + \sigma_{xy}^2}, \quad (2.6)$$

$$\rho_{xy} = \frac{\sigma_{xy}}{\sigma_{xx}^2 + \sigma_{xy}^2}. \quad (2.7)$$

In the case of the classical Hall effect ρ_{xx} and ρ_{xy} can be read off from (2.3)

$$\rho_{xx} = \frac{m^*}{e^2 n \tau}, \quad (2.8)$$

$$\rho_{xy} = \frac{B}{ne}. \quad (2.9)$$

The transverse resistivity is also called the Hall resistivity. From (2.8), it follows that there is no magnetoresistivity in the longitudinal direction [3].

Using (2.8) and (2.9) expressions for the charge carrier density n and the carrier mobility μ can be obtained. The charge carrier density arises in the classical explanation of the Hall effect. It is therefore ideally extracted at $B = 0$, where the system is the least quantum mechanical. In chapter 3 it will be discussed how the magnetic field plays a role in the quantum theory. In order to get a useful result, the slope of ρ_{xy} evaluated at $B = 0$ is used. If not, the expression would involve either multiplication or division by zero. By differentiating (2.9) with respect to B the following expression for n is obtained

$$n = \left(e \left. \frac{d\rho_{xy}}{dB} \right|_{B=0} \right)^{-1}. \quad (2.10)$$

The mobility μ is the proportionality constant between the drift velocity and the electric field at zero magnetic field. It is also defined as [5]

$$\mu = \frac{e\tau}{m^*}. \quad (2.11)$$

Substituting this into (2.8) one obtains the expression

$$\mu = \frac{1}{en\rho_{xx}|_{B=0}}. \quad (2.12)$$

Usually, the resistance R differs from the resistivity ρ due to geometric factors. This is also the case for the longitudinal resistivity which can be calculated with [3]

$$\rho_{xx} = \frac{U_{xx}}{I} \frac{W}{L} = R_{xx} \frac{W}{L}. \quad (2.13)$$

However, for the transverse resistivity this is not the case as it can be calculated with [3]

$$\rho_{xy} = \frac{U_{xy}}{I} = R_{xy}. \quad (2.14)$$

This outcome is convenient since the variable of interest, ρ_{xy} , coincides with the measured value, R_{xy} , in this particular case. On the other hand, when measuring the longitudinal resistance R_{xx} , one must multiply by the appropriate width to length ratio to obtain the resistivity ρ_{xx} .

2.1 Limitations of the classical model

Looking at (2.9), the classical Hall effect predicts a linear increase in transverse resistivity with increasing magnetic field. However, this behavior breaks down for a two-dimensional system at low temperatures and high magnetic fields, where quantum effects dominate. When measuring ρ_{xy} under these conditions, the following is observed:

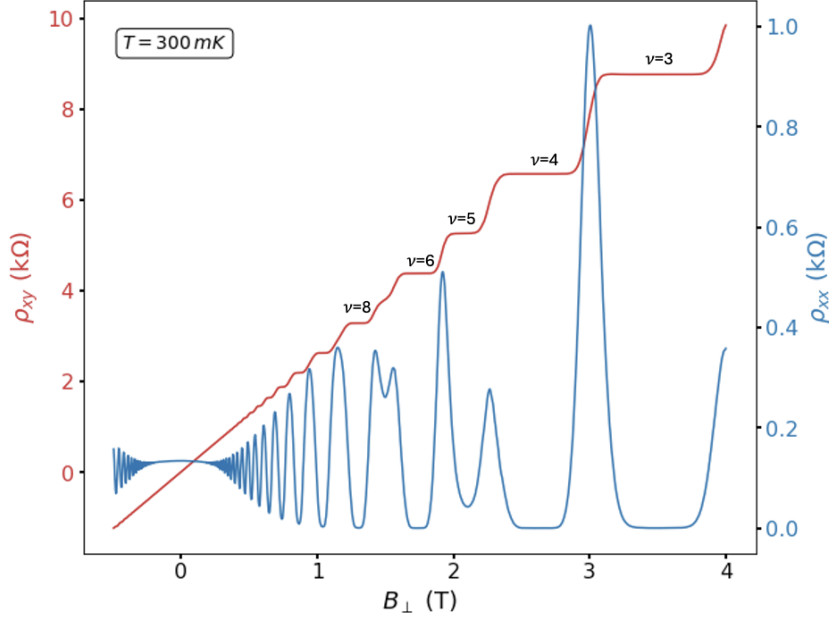


Figure 2.2: transverse resistivity measurement at large magnetic field and low temperature. Performed at QTL.

The phenomenon observed above is known as the quantum Hall effect. The resistivity plateaus at values of $e^2/h\nu$, where ν has been dubbed the filling factor. When first discovered ν was measured to take on integer values, and this effect has therefore been dubbed the integer quantum Hall effect (IQHE). Later experiments found plateaus that exist at fractional values of ν , and this effect has appropriately been named the fractional quantum Hall effect. This project focuses on the IQHE, as it appears in the experiment conducted by the students. Going forward I will therefore just refer to the IQHE as the QHE. In the next two chapters a quantum mechanical approach will be used to explain the plateaus.

Before the plateaus start forming in ρ_{xy} , oscillations can be seen in ρ_{xx} . These are called Shubnikov-de Haas (SdH) oscillations. The theory behind them won't be elaborated in the main body of this project, but appendix B is dedicated to explaining them, as they will be used during the data analysis in chapter 7.

3 Landau levels

The setup is now revisited from a quantum mechanical point of view. It is assumed that the two-dimensional system extends in the xy -plane. The goal is now to find wave functions subject to a magnetic field in the z -direction $\mathbf{B} = (0, 0, B)$. In order to find these, the time-independent Schrödinger's equation $H\psi = E\psi$ is solved. This is done in the Landau gauge $\mathbf{A} = (-By, 0, 0)$ [3]. The Hamiltonian takes the form

$$\hat{H} = \frac{1}{2m^*}(\mathbf{p} + |e|\mathbf{A})^2 = \frac{1}{2m^*}((\hat{p}_x - |e|By)^2 + \hat{p}_y^2), \quad (3.1)$$

where e is the elementary charge.

As seen in the Hamiltonian there is translation invariance along the x -axis. Therefore, the eigenstates of (3.1) must also be eigenstates of $\hat{p}_x$ [6]. That is, functions where $\hat{p}_x f(x) = \hbar k_x f(x)$, with $\hbar$ being Planck's constant. These are just plane waves $e^{ik_x x}$. This allows for the following ansatz

$$\psi_k(x, y) = e^{ik_x x} f_{k_x}(y). \quad (3.2)$$

When the Hamiltonian acts on this wave function the operator $\hat{p}_x$ returns its eigenvalue $\hbar k_x$. The LHS of the Schrödinger equation can now be written as

$$\hat{H}\psi_k(x, y) = \frac{1}{2m^*}((\hbar k_x - |e|By)^2 + \hat{p}_y^2)\psi_k(x, y). \quad (3.3)$$

The term $\frac{1}{2m^*}(\hbar k_x - |e|By)^2$ can be rewritten by introducing the cyclotron frequency $\omega_B = \frac{|e|B}{m^*}$ and the magnetic length $l_B = \sqrt{\frac{\hbar}{|e|B}}$ [6]

$$\begin{aligned} \frac{1}{2m^*}(\hbar k_x - |e|By)^2 &= \frac{1}{2m^*}(|e|B)^2 \left(y - \frac{\hbar k_x}{|e|B}\right)^2 \\ &= \frac{1m^*}{2(m^*)^2}(|e|B)^2(y - k_x l_B^2)^2 = \frac{1}{2}m^*\omega_B^2(y - k_x l_B^2)^2. \end{aligned} \quad (3.4)$$

Following this the Schrödinger equation takes the form:

$$\left(\frac{\hat{p}_y^2}{2m^*} + \frac{1}{2}m^*\omega_B^2(y - k_x l_B^2)^2\right)\psi_{k_x}(x, y) = E\psi_{k_x}(x, y). \quad (3.5)$$

This is a 1D harmonic oscillator along the y -axis with its center moved $k_x l_B^2$ from the origin [6]. The momentum in the x -direction, $\hbar k_x$, has moved the position of the harmonic oscillator in the y -direction, so that it is now centered about $y_0 = k_x l_B^2$. The energy spectrum is

$$E_n = \hbar\omega_B \left(n + \frac{1}{2}\right), n \in \mathbb{N}, \quad (3.6)$$

corresponding to discrete energy levels known as Landau levels [6]. For a fixed value of n , the energies are degenerate for different values of k_x . In appendix C the degeneracy of each Landau level is calculated.

The wave functions are given by the product of the plane wave and the 1D harmonic oscillator wave functions:

$$\psi_{n,k_x}(x, y) \sim e^{ik_x x} H_n \left(\frac{y - k_x l_B^2}{l_B^2}\right) e^{-(y - k_x l_B^2)^2 / 2l_B^2}, \quad (3.7)$$

with H_n being the Hermite polynomials [6]. The $\sim$ indicates that the wave functions haven't been normalized. Part of the wave function is a Gaussian that localizes the eigenfunction around $y = k_x l_B^2$, while it is completely extended in the x -direction.

The sample used in QHE experiments is finite and has edges. The wave functions are localized in one direction, so there should be states that live near the edge of the sample [7]. These states are appropriately called edge states and play an important role in the understanding of the plateaus seen in the experimental data of ρ_{xy} in fig. 2.2. One could say that the key to understanding the formation of plateaus lies at the edges of the sample.

4 The quantum Hall effect

4.1 Edge modes

Looking at the edges from a classical perspective, it is clear that something special happens. Each particle in a fixed magnetic field is moving in the same orbital motion, for instance anti-clockwise. Near the edge of the sample, the orbits collide with the boundary. As all motion is anti-clockwise they bounce back resulting in a skipping motion. The particles along the one-dimensional boundary move only in a single direction. For the two sides of the sample they move in opposite directions.

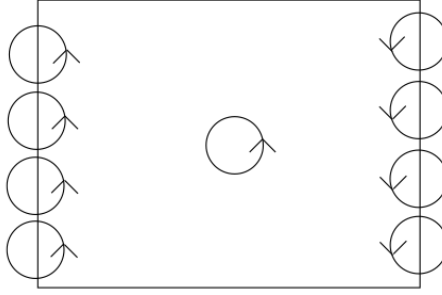


Figure 4.1: Skipping orbit of particles along the two edges of the sample [6].

The edge modes also appear in the quantum theory. The edge of the sample is modeled by a potential $V(y)$ which rises steeply as seen in fig. 4.2a. Working in the Landau gauge and considering this potential the Hamiltonian is

$$\hat{H} = \frac{\hat{p}_y^2}{2m^*} + \frac{1}{2}m^*\omega_B^2(y - k_x l_B^2)^2 + V(y). \quad (4.1)$$

Since the states of the system are localized in the y -direction, and by assuming that the potential varies slowly on the scale of the magnetic length, the edge potential $V(y)$ can be expanded around the location $y = Y$. Each wave function then experiences the potential $V(y) \approx V(Y) + (\partial V / \partial y)(y - Y) + \dots$ [7]. Dropping quadratic and constant terms and plugging the potential back into (4.1) gives

$$\hat{H} = \frac{\hat{p}_y^2}{2m^*} + \frac{1}{2}m^*\omega_B^2(y - k_x l_B^2)^2 + y \frac{\partial V}{\partial y}. \quad (4.2)$$

This Hamiltonian can be rewritten to get another displaced harmonic oscillator plus some extra terms that come from completing the square. The resulting energy spectrum is [7]

$$E_{n,k_x} = \hbar\omega_B \left(n + \frac{1}{2} \right) - \frac{1}{2m^*\omega_B^2} \left(\frac{\partial V}{\partial y} \right)^2 + k_x l_B^2 \frac{\partial V}{\partial y}. \quad (4.3)$$

This spectrum depends on the x -momentum $\hbar k_x$, and so it gives rise to a non-zero group velocity in the x -direction at the edges given by [7]

$$v_x = \frac{1}{\hbar} \frac{\partial E_{n,k_x}}{\partial k_x} = \frac{1}{\hbar} \frac{\partial V}{\partial y} l_B^2 = \frac{1}{|e|B} \frac{\partial V}{\partial y}. \quad (4.4)$$

On the left side of the sample $(\partial V/\partial y) < 0$ so here $v_x < 0$. The opposite applies on the right side resulting in $v_x > 0$. The modes at each edge travel in opposite direction. This fits with the classical explanation of skipping orbits.

By introducing a Fermi level, available states can be filled in. The states are labeled by x -momentum $\hbar k_x$, but this can equally be thought of as the position of the state in the y -direction [6]. This is due to the wave functions being localized around $y = k_x l_B^2$, which was explained in chapter 3. The filled states can then be drawn in as seen in fig. 4.2b.

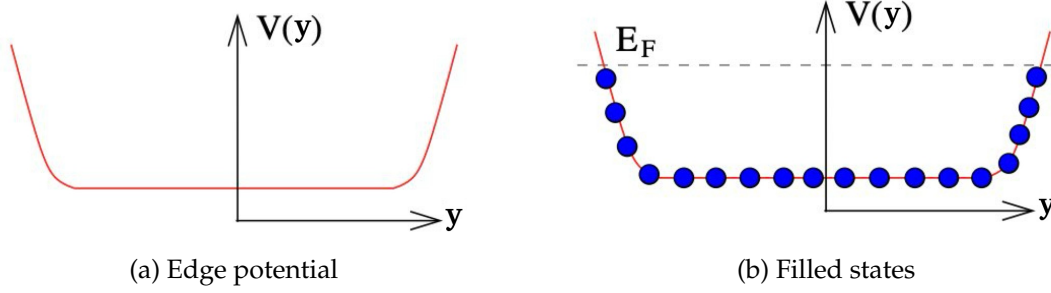


Figure 4.2: Empty and filled edge modes [6].

In order to understand why the plateaus take the famous value $\rho_{xy} = h/e^2\nu$ the current along the edges I_x can be studied. To study this, a potential difference ΔU is introduced between the two edges of the sample. In an experimental setup, this arises naturally due to the Lorentz force pushing the electrons to one side of the sample. This means that more states are filled up on one edge of the sample.

It is possible to calculate the arising current along the edges I_x caused by the potential difference ΔU

$$I_x = -|e| \int \frac{dk_x}{2\pi} v_x = \frac{-|e|}{2\pi l_B^2} \int dy \frac{1}{|e|B} \frac{\partial V}{\partial y} = \frac{-|e|}{h} \int dy \frac{\partial V}{\partial y} = \frac{|e|}{h} \Delta U, \quad (4.5)$$

where the integration was done over one completely filled Landau level [6]. By noting that the transverse voltage across the sample is given by the potential difference $U_{xy} = \Delta U/|e|$, the transverse conductivity can be calculated [6]

$$\sigma_{xy} = \frac{I_x}{U_{xy}} = \frac{e^2}{h}. \quad (4.6)$$

This is the conductivity for a single Landau level. For ν filled levels it takes the value $\sigma_{xy} = \nu e^2/h$. Remembering that $\rho_{xx} = 0$ when the plateaus are present, (2.7) can be used to calculate the resistivity of the plateaus:

$$\rho_{xy} = \frac{1}{\nu} \frac{h}{e^2}. \quad (4.7)$$

It is possible to understand the formation of plateaus by examining the edge-potential while available states are filled. Generalizing the filling of available states in fig. 4.2b to several Landau levels fig. 4.3 is obtained.

All Landau levels with energy below the Fermi level are filled, while those above are empty. Changing the magnetic field alters the spacing between Landau levels, as seen from the energy $E_n = \hbar\omega_B(n + 1/2)$. As long as the Fermi level lies between two Landau levels, the number of filled edge modes remains unchanged. Since each Landau level is associated with one edge mode, the number of filled Landau levels directly corresponds to the number

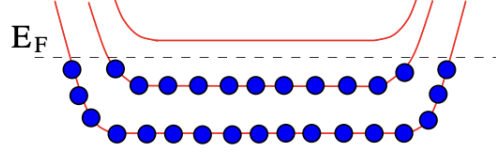


Figure 4.3: Edge modes [6].

of conducting edge states. Therefore, ρ_{xy} remains constant as the number of available edge states does not change until the increased spacing between Landau levels moves an edge mode above the Fermi level, forming the plateaus [6].

4.2 The role of disorder

So far ideal, disorder-free systems like the potential in fig. 4.2a have been studied. However, in realistic samples, the potential has fluctuations due to imperfections in the material, as indicated on fig. 4.4. These impurities play a critical role in the formation of plateaus.

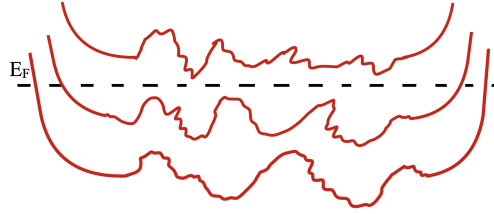


Figure 4.4: Edge modes with disorder

These impurities cause electrons in the bulk of the material to get trapped in localized orbits. This localization plays a crucial role in the quantization of transverse resistivity. When the Fermi level lies between two Landau levels, the extended bulk states are absent, and all remaining states in the bulk become localized and do not contribute to transport. As a result, only edge states, which remain conducting, carry current. This leads directly to the formation of quantized Hall plateaus.

The disorder must not be too large. If so, the localized states in different Landau levels can start to overlap, and some of them might become extended again. This would disrupt the quantization of transverse resistivity. For the QHE to work properly, you need just the right amount of disorder; enough to localize the bulk states, but not so much that it destroys the clear separation between Landau levels.

5 Observing and measuring the QHE

In order to observe and measure the QHE you need an advanced experimental setup. A two-dimensional electron gas (2DEG) is required, typically created in semiconductor heterostructures such as GaAs/AlGaAs. Highly sensitive instrumentation is necessary to detect the discrete plateaus in the Hall resistance ρ_{xy} and the vanishing longitudinal resistance ρ_{xx} .

A strong magnetic field and low temperatures are required because the thermal broadening of the Landau levels, given by $k_B T$, where k_B is the Boltzmann constant, must be smaller than the Landau level spacing $\hbar\omega_B$ [3]. A clean sample is also essential because, as discussed in section 4.2, the broadening of edge modes caused by impurities also must not exceed the Landau level spacing [3]. Modeling the broadening caused by impurities as a random poten-

tial V , these two requirements can be compactly written as

$$\hbar\omega_B > k_B T + V. \quad (5.1)$$

5.1 Two-dimensional electron gas

A 2DEG is a system where electrons are free to move in two dimensions, typically x and y , but are tightly confined in the third dimension, in this case z . This quantum confinement restricts the motion in the z -direction, leading to quantized energy levels called subbands, while allowing free movement in the plane. This creates a two-dimensional layer of conducting electrons [8].

It is possible to produce a 2DEG by placing a semiconductor, such as GaAs with a band gap of 1.42 eV , between two semiconductors with a greater band gap, such as AlGaAs. When adding Aluminum to GaAs the band gap increases from 1.42 eV up to 2.16 eV (going from pure GaAs to pure AlAs) [9]. This structure constructed by stacking two or more different semiconductor materials together is called a heterostructure. A heterostructure can be seen in fig. 5.1. When a layer of GaAs is sandwiched between AlGaAs layers, a quantum well is formed due to the conduction band offset between the two materials [8]. If the well is constructed such that the relation between the chemical potential E_F and the two lowest subbands E_1 and E_2 is $E_1 < E_F < E_2$, the electrons are completely confined to the well.

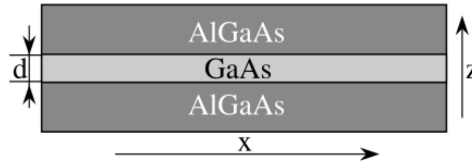


Figure 5.1: A heterostructure made from GaAs and AlGaAs [10].

A schematic of a quantum well can be seen in fig. 5.2. For this configuration $E_{C_{\text{AlGaAs}}} > E_{C_{\text{GaAs}}}$ and $E_{V_{\text{AlGaAs}}} < E_{V_{\text{GaAs}}}$. This creates an energy barrier between the two semiconductors, creating the quantum well in the direction of the heterostructure [8].

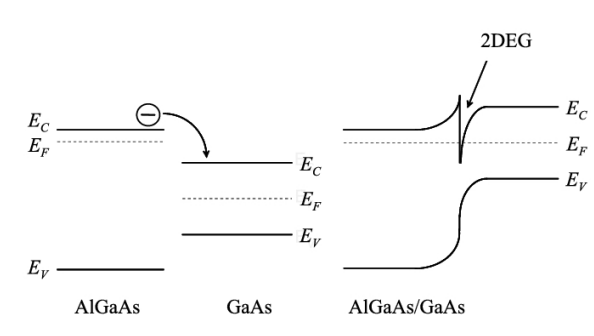


Figure 5.2: Change in potential after combining the two materials [11].

5.2 4-wire resistance measurements

It is not possible to directly measure the resistance of a component in a circuit. When performing the measurement, a small current is sent through the component, and the voltage drop across it is measured. Then Ohm's law $R = U/I$ is used to calculate the resistance [12]. When measuring the resistance of a component with an ohmmeter, a two-wire measurement is performed. A schematic of this can be seen in fig. 5.3. The resistance in the wires connecting the component with the current generator causes a voltage drop U_w . Since the voltage meter

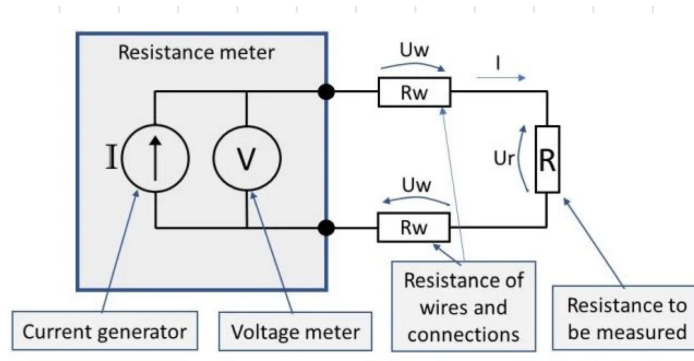


Figure 5.3: Two-wire measurement [12].

is connected to the same two wires, the measured voltage drop is $U_r + 2U_w$. The measured voltage drop used for calculating the resistance is increased, and the calculated resistance is therefore greater than the actual resistance of the component. In order to improve resistance measurements, a four-wire setup like the one in fig. 5.4 can be utilized. Using four wires the

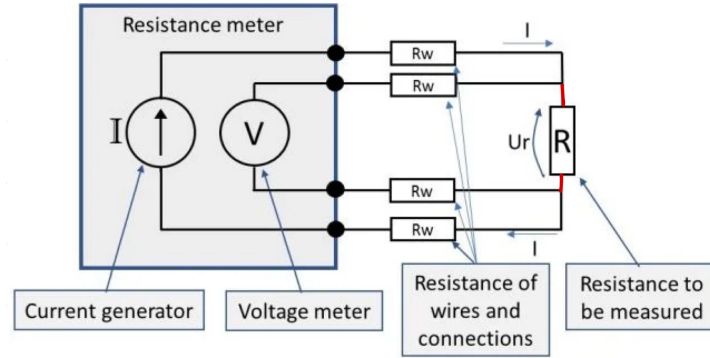


Figure 5.4: Four-wire measurement [12].

current generator and the voltage meter are connected to the component with two different pairs of wires. There are still voltage drops across the two wires connected to the current generator, but they don't get measured by the voltage meter. The impedance of the voltage meter is large enough so that virtually no current is running across it [12]. Thus, the voltage drop across that set of wires is close to zero.

It's now possible to get an accurate measurement of both the voltage drop across the component and the current going through it. This setup allows for measuring the resistance of a component more accurately. The red part of the wire still has a voltage drop, so the measured resistance will still be greater than the actual resistance, but this has been significantly reduced. This method is essential for getting accurate measurements of the QHE, since any parasitic resistance can distort the measurements of the quantized ρ_{xy} and the vanishing ρ_{xx} .

5.3 Adiabatic demagnetization refrigeration (ADR)

ADR is a technology used in Kiutra cryostats to achieve near zero temperatures. A reservoir R cools a cooling medium and the sample to a temperature of 4 K. The cooling medium, a magnetocaloric material, becomes magnetized. A magnetocaloric material warms up when a magnetic field is applied [13], but due to contact with the reservoir R the temperature is constant at 4 K. After the cooling the heat switch is closed and the medium gets demagnetized cooling it and the sample down below 4 K to the target temperature needed to carry out the experiment. Since the system cannot be completely isolated, the temperature will begin to rise to that of its surroundings. In order to combat this, the magnet slowly demagnetizes

completely. This means that there is a limit to the length of time the sample can remain at the target temperature. This is called hold time and is typically 4-6 hours for the cryostat used in this project [14].

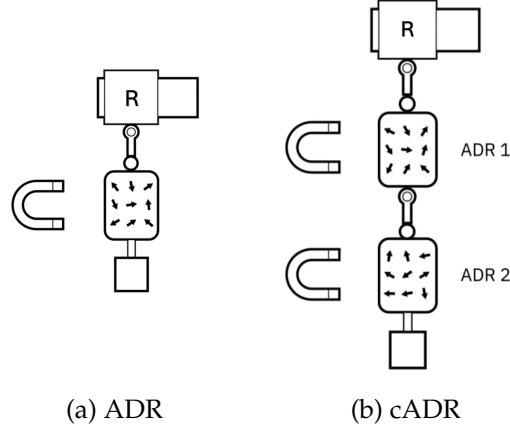


Figure 5.5: Schematics for the two different cooling modes ADR and cADR [14].

5.3.1 Continuous ADR (cADR)

In order to continuously keep the sample at the target temperature, a chain of two ADR's can be set up. The two cooling mediums take turns being demagnetized. While one demagnetizes, the other gets magnetized again. Whenever one is completely demagnetized, the other starts demagnetizing, ensuring the temperature is kept at the target temperature. This can go on indefinitely [14].

5.4 Lock-in Amplifier

Lock-in amplifiers are used to measure small AC signals all the way down to a magnitude of a few nanovolts. Precise measurements can be made even when the small signal is blurred by noise sources several orders of magnitude greater [15].

The amplifier works by multiplying the incoming signal $U_{sig}\sin(\omega_r t + \theta_{sig})$ with its reference signal $U_L\sin(\omega_L t + \theta_{ref})$ using a phase-sensitive detector (PSD) [15]. The output is

$$\begin{aligned} U_{psd} &= U_{sig}U_L\sin(\omega_r t + \theta_{sig})\sin(\omega_L t + \theta_{ref}) \\ &= \frac{1}{2}U_{sig}U_L\cos((\omega_r - \omega_L)t + \theta_{sig} - \theta_{ref}) \\ &\quad - \frac{1}{2}U_{sig}U_L\cos((\omega_r + \omega_L)t + \theta_{sig} + \theta_{ref}). \end{aligned} \quad (5.2)$$

If this output is sent through a low pass filter both terms die, since they are high frequency AC. If $\omega_L = \omega_r$ then

$$U_{psd} = \frac{1}{2}U_{sig}U_L\cos(\theta_{sig} - \theta_{ref}) - \frac{1}{2}U_{sig}U_L\cos((\omega_r + \omega_L)t + \theta_{sig} + \theta_{ref}). \quad (5.3)$$

The difference frequency term becomes a DC signal and if the output is then sent through a low pass filter the difference term survives while the sum term dies out. The final output after passing through the low pass filter is

$$U_{psd} = \frac{1}{2}U_{sig}U_L\cos(\theta_{sig} - \theta_{ref}). \quad (5.4)$$

The combination of a PSD and a low pass filter effectively removes noise, because neither $\omega_{noise} + \omega_{ref}$ nor $\omega_{noise} - \omega_{ref}$ is close to DC.

Only signals at frequency $\omega = \omega_L$ survive. To ensure correct detection, the reference phase must be locked so that the phase difference $\theta = \theta_{sig} - \theta_{ref}$ remains constant. The PSD output is proportional to $U_{sig} \cos(\theta)$. The phase dependency of the measurement can be eliminated if another PSD is used. For this PSD the reference signal is $U_L \sin(\omega_L t + \theta_{ref} + 90^\circ)$. Sending the signal through the low pass filter results in the output

$$U_{psd2} = \frac{1}{2} U_{sig} U_L \cos(\theta_{sig} - \theta_{ref} - 90^\circ) = \frac{1}{2} U_{sig} U_L \sin(\theta_{sig} - \theta_{ref}). \quad (5.5)$$

Output 1 is labeled X and output 2 Y. Then $X = U_{sig} \cos(\theta)$ and $Y = U_{sig} \sin(\theta)$ [15]. The size of this signal is $R = \sqrt{X^2 + Y^2} = U_{sig}$. A dual-phase lock-in can measure X, Y, and R directly. The phase is calculated using the following formula

$$\theta = \arctan\left(\frac{Y}{X}\right). \quad (5.6)$$

5.5 Setting up the student experiment

The student experiment is performed on a Kiutra cryostat. The single ADR mode was used for the QHE experiment as it took less than 6 hours to perform it. A Hall bar device fabricated on a GaAs/AlGaAs heterostructure is loaded into the cryostat. In order to measure the signal across the Hall bar, three lock-ins are connected to the chip with BNC cables that are hooked up to a breakout box. For this experiment SR-860 lock-ins are used. The master lock-in applies a source voltage of 100 mV that passes through a load resistor of $1 \text{ M}\Omega$. The resistance of the bias circuit is defined by the load resistor since the sample resistance is on the scale of 100Ω . This results in a measured bias current of approximately 100 nA . The other two lock-ins measure U_{xx} and U_{xy} accordingly. The lock-ins are also connected to a computer. Using a program, the computer can make the cryostat perform a sweep of the magnetic field while collecting the data from the lock-ins. A drawing of the setup can be seen in fig. 5.6.

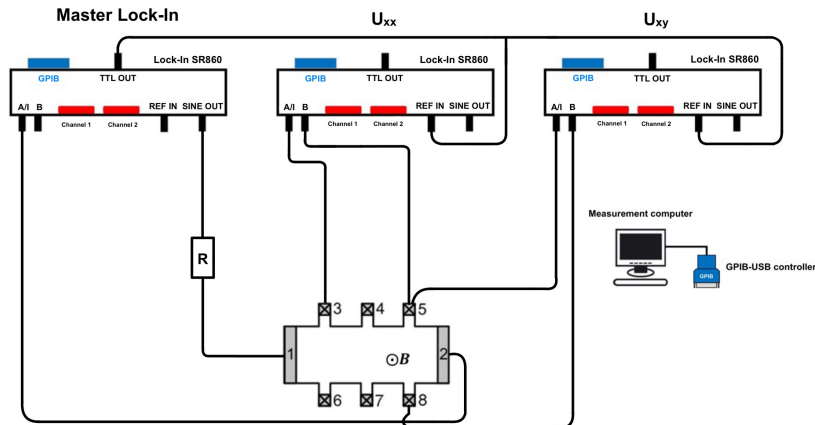


Figure 5.6: Schematic of the experimental setup used in the student measurement. The master lock-in provides the source voltage and measures current, while the two additional lock-ins measure the longitudinal voltage U_{xx} and transverse voltage U_{xy} .

During the student experiment two measurements were made. The first with $T = 3 \text{ K}$ and the second with $T = 300 \text{ mK}$. For the first measurement the magnetic field was swept from -0.5 T to 4 T , while for the second, a reverse sweep from 4 T to -0.5 T was used to save

time. For both magnetic field sweeps a ramping speed of 0.2 T/min was used, and the lock-ins were set at a frequency of 17 Hz .

6 Results from observing and interviewing students

	Perception of the experiment	Acquired understanding and knowledge	Acquired practical laboratory skills
Observations	Mainly engaged by questions from the teacher	Discussions with teacher about theory behind experiment	- Few tasks for the students - Experiment was mostly carried out by a computer
Interview	- Appreciation of experiencing quantum mechanics (>1 student) - Fascination about state-of-the-arts equipment (1 student)	- Distinction between classical and quantum mechanics (1 student) - Clearer understanding of QHE (1 student)	Prepared for reproducing experiment (1 student)
Both obs. & int.	- Waiting time is not experienced as a problem (>1 student) - Collaboration between students and the teacher (>1 student)	- Experience with QHE (>1 student) - Experience with Zeeman splitting and SdH oscillations (1 student)	- Few experimental skills (1 student) - Experience in testing theories experimentally (1 student)

Figure 6.1: The image above presents the key findings from observing and interviewing a group of four students performing the QHE experiment. Each column represents a category of findings, while each row indicates whether the findings are based on observation, interview, or both. The same four students were both observed and interviewed. For findings based on interviews, the number of students who either agreed with or repeated a given statement is shown. This is further divided into responses from one student or more than one student.

The main goal of the observation and interview was to explore how the students perceived the experiment, whether they acquired any understanding regarding the physics involved in the experiment, or acquired any practical laboratory skills. In addition to myself, a fellow student also observed the experiment, which allowed us to compare our observations afterwards. Before the observation I wrote a form making it easier to categorize observations along the way. A segment of the form can be seen in fig. 6.2. I also wrote a list of interview questions beforehand.

The interview was done in Danish. Since this report is written in English, I have translated the answers to the best of my abilities. Do keep in mind that translations can never truly capture the meaning of the original answer. The original answers are not included in this project, but they are available upon request.

6.1 Perception of the experiment

The experiment involved few hands-on tasks, focusing instead on teacher-led questions to engage students.

The students appreciated applying their quantum mechanics knowledge and were ex-

cited to use advanced equipment, especially working at 300 mK. *"So this is the first experiment where I think it makes sense in relation to quantum mechanics to understand why this is quantum, and what makes it special and unique, with trying to see the Hall bar and all these other things". "That's cool! It's pretty cool to be able to work with a cryostat. Holy s**t man, 0.3 Kelvin".*

Although some parts of the experiment were time-consuming with little active work, students remained engaged through interaction with the teacher. Rather than a typical group setup, the experience felt more like a collaborative effort between students and teacher. *"I felt like it was very much like everyone collaborated with [...] [the lecturer] in some way. Because I don't know if... It wasn't really a group experiment where all four of us had some kind of job to do like that". "It was more of a conversation between [...] [the lecturer] and us, wasn't it? He asked a lot about what we thought about the experiment and our expectations and what we thought about our results. And it wasn't like in some of the other experiments we've tried, where the teacher or lecturer is looking for a very specific answer".*

6.2 Acquired understanding and knowledge

Throughout the experiment, the teacher discussed relevant theory with the students; initially to guide the procedure, and later to help interpret results.

The experiment deepened their understanding of the distinction between classical and quan-

Activity	Focal points	Observations
B-sweep at 300 mK	- What do the students do while waiting? - Do they analyze the data while it is being collected?	- The students follow the live data as the teacher explains theoretical aspects of the observations. - The students ask questions about the cooling process.

Figure 6.2: Segment of observation form

tum mechanics. It also clarified how quantum effects emerge at low temperatures and strong magnetic fields. *"I would say it [the experiment] shows how quantization of things gets weird and behaves differently than classical. When you start getting down into these quantum levels, like with very cold temperatures and high magnetic fields and things like that, and looking at these discrete levels".*

The students now better understand the quantum Hall effect and related phenomena like Zeeman splitting and SdH oscillations. *"I now understand much more how quantum hall works in general. I think it has given me a slightly better intuition about how, for example, the size of magnetic fields will affect how electrons move". "I think it was cool to see the Zeeman effect in those Shubnikov-de Haas oscillations. I think it was... I didn't expect that actually. I just expected to see the oscillations and then without the Zeeman effect. So there I was like, okay, it's actually cool that you can see that in our measurements too".*

6.3 Acquired practical laboratory skills

The experiment required few manual tasks beyond connecting cables and running a Python program that controlled the sweep, cooling, and data collection.

Despite the limited hands-on work, students feel well-equipped to conduct future experiments involving the quantum Hall effect. *"If I were to do some experiment where I had to use the quantum Hall effect, for example. If I were to make some kind of micro insulator, I would probably be able to use the principles from this experiment much more easily for that".*

While tangible experimental skills were minimal, the students gained valuable experience in setting up experiments to explore and test physical theories. *"So, the experiment itself, I think, gave a pretty fun insight into how you actually test for these things. So, what kind of... So, how you set up a circuit to test for something or other, more like... So, it's getting very vague now, but like... How you set up experiments to test physical theories"*.

7 Further analysis of QHE data

During the project I was in the laboratory and performed the QHE experiment twice. The first time I performed the same measurements as the students. This gave me a better understanding of the experiment and enabled me to ask more informed questions during the interview. After the experiment the students were tasked to hand in a report. Some of the requirements for the report are mentioned in appendix D.

I went further than the students in my analysis of the data, extracting additional values allowing me to characterize the Hall bar sample in greater detail. This involved measuring the effective mass and calculating the mean free path and scattering time.

During the first experiment I started by performing a sweep from -0.5 T to 4 T at 3 K and then again at 300 mK in both cases looking at the QHE in the transverse resistivity ρ_{xy} . The second time I instead examined SdH oscillations. Here I performed 5 measurements from -0.1 T to 1.3 T at temperatures from 300 mK to 700 mK with iterations of 100 mK .

7.1 Calculating mean free path and effective mass

In this section a brief explanation of how to calculate the mean free path l_e and the effective mass m^* will be presented. The mean free path is the average distance between scattering events [3]. A particle's effective mass is defined as being the quantity that satisfies Newton's second law, $F = m^*a$, for the particle in question [16].

It is possible to calculate the mean free path of the electrons in the sample using the formula [5]

$$l_e = \frac{\mu\hbar\sqrt{2\pi n}}{e}. \quad (7.1)$$

The amplitude of the SdH oscillations are reduced by increasing temperature [5]

$$A(T, B_m) = 2e^{-\frac{\pi}{\omega_B \tau_q}} \frac{2\pi^2 k_B T m^* / \hbar |e| B_m}{\sinh(2\pi^2 k_B T m^* / \hbar |e| B_m)}, \quad (7.2)$$

where B_m is the magnetic field at the position of the m -th peak of the SdH oscillations. The relative amplitude between temperature T and base temperature T_0 is

$$\frac{A(T, B_m)}{A(T_0, B_m)} = \frac{T \sinh(2\pi^2 k_B T_0 m^* / \hbar |e| B_m)}{T_0 \sinh(2\pi^2 k_B T m^* / \hbar |e| B_m)}. \quad (7.3)$$

By measuring and plotting the relative amplitude versus T , equation (7.3) can be used to fit the data with m^* being the only fit parameter [5].

7.2 Setting up the experiment

In section 5.5 I wrote an explanation of the experimental setup that the students used. I used an identical setup the first time I performed the QHE experiment. An illustration of the setup can be seen in fig. 5.6.

For the second experiment, the setup was nearly identical, except that I used a load resis-

tor of $10\text{ M}\Omega$ and a source voltage of 100 mV resulting in a measured bias current of approximately 10 nA . The lock-ins were again set at a frequency of 17 Hz . In appendix E I elaborate on the ramping speeds used for the magnetic field in this experiment.

7.3 Presentation of results

Fig. 2.2 shows my results from the measurement at $T = 300\text{ mK}$ from the first experiment I performed. In fig. 7.1 the data from my second measurement can be seen. The measurements were performed with varying ramping speeds due to unforeseen program behavior, as discussed in appendix E.

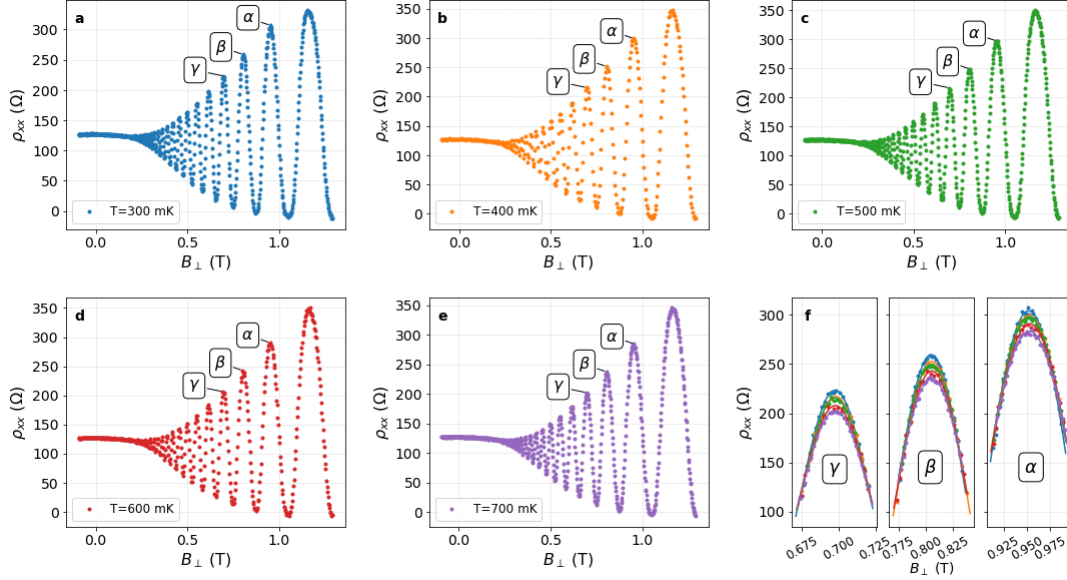


Figure 7.1: Plots a - e show measurements of ρ_{xx} at different temperatures, with color coding consistent across all plots. Plot f shows local sine curve fits to peaks α , β , and γ used to extract the temperature-dependent amplitude of the SdH oscillations.

Plot f of fig. 7.1 shows sine curves fitted locally around peaks of the SdH oscillations. For each peak the amplitude of the fitted sine curve was recorded and then used in order to extract the effective mass using (7.3). Fitting three different peaks yielded three independent estimates of the effective mass m^* , which can all be seen in fig. 7.2. Since I obtained three

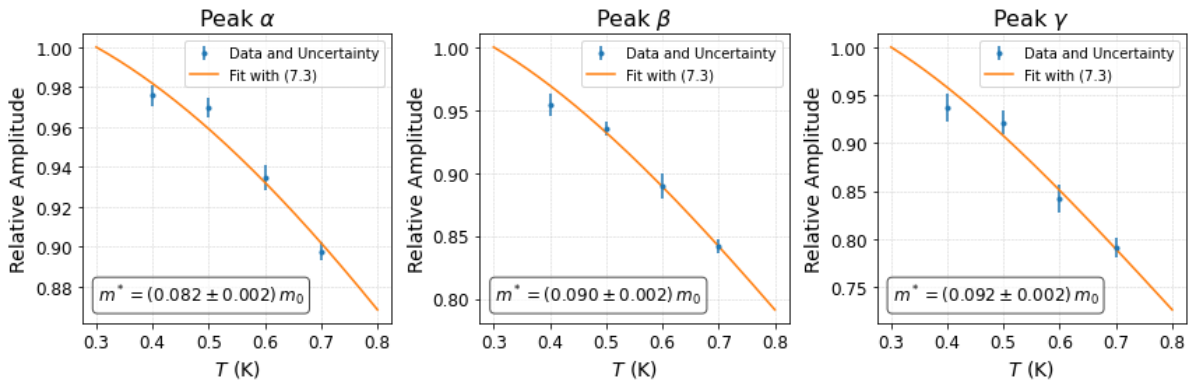


Figure 7.2: Fit of the relative amplitude of SdH oscillations as a function of temperature, used to extract the effective mass m^* . Error bars correspond to uncertainties derived from the square root of the diagonal of the covariance matrix.

different values for the effective mass, I calculated the weighted mean. The method used for this as well as additional plots of the individual fits to each peak can be seen in appendix E.

There, I also explain propagation of uncertainty.

Using data from the first experiment (QHE sweep at 3 K and 300 mK), I extracted the electron density and mobility. Since the expressions, (2.12) and (2.10), are derived from the theory of the classical Hall effect, the data from measuring at $T = 3$ K gives more accurate results. At this temperature the electrons behave less quantum mechanically.

I then used the data from the second experiment (SdH oscillations at multiple temperatures) to calculate the effective mass of the electrons with the method explained in this section. Having extracted it I was then able to calculate the scattering time of the sample using (2.11) [3].

In table 7.1 I have written my extracted values for electron density, mobility, mean free path, effective mass, and scattering time.

Table 7.1: Summary of extracted parameters from the QHE experiments. The listed uncertainties are calculated by taking the square root of the diagonal of the covariance matrix and/or from propagation of uncertainty

Parameter	Symbol	Value	Unit	Source
Electron density	n	$(2.4762 \pm 0.0005) \times 10^{15}$	m^{-2}	QHE data (3 K)
Mobility	μ	$(18.758 \pm 0.004) \times 10^4$	$\text{cm}^2/\text{V}\cdot\text{s}$	QHE data (3 K)
Mean free path	l_e	(1.5401 ± 0.0004)	μm	Derived from μ, n
Effective mass	m^*	$(0.088 \pm 0.001) m_0^a$	—	SdH data (300–700 mK)
Scattering time	τ	(9.4 ± 0.1)	ps	Derived using m^*

^a Here, $m_0 = 9.109 \times 10^{-31}$ kg denotes the rest mass of the electron.

7.4 Discussion of results

This will be a brief discussion of the extracted value of the effective mass m^* . GaAs has an effective mass of $0.067 m_0$ [9], while AlAs has a value of $0.15 m_0$ [9]. My measured effective mass of $(0.088 \pm 0.001) m_0$ lies between these, suggesting that the electron wavefunction is partially confined to both materials. Since it is closer to the value of GaAs, this suggests that it is predominantly localized in the GaAs layer.

8 Ideas for future implementations of the QHE experiment

Overall, I think the implementation of the QHE experiment and the use of the QTL within the Nano3 course worked very well. The students seemed to get a solid grasp of the theory behind the QHE, and they stayed engaged throughout the experiment. The setup also allowed for a lot of good conversations between the students and the lecturer. That said, there are definitely a few things I would consider changing or improving; especially if the experiment is to be scaled up for larger courses.

In the current format, each group performs the experiment one at a time, with the lecturer acting as the TA. This works fine in Nano3, which only has 21 students enrolled. However, it is not practical for larger courses, such as the compulsory undergraduate courses for physics majors, which can have upwards of 100 students.

One of the major bottlenecks is simply time. The cryostat has a fixed speed for sweeping the magnetic field and adjusting the temperature, which places a lower limit on how quickly each experiment can be run. Currently, the QTL only has one cryostat available, so there's no realistic way to increase throughput without acquiring additional equipment. That's a clear logistical limitation that would need to be addressed before expanding the experiment to a

broader course offering.

In terms of workflow, the experiment involves quite a bit of downtime while measurements are running. In Nano3, that time is filled with discussions between the students and the TA. While this works well in a small course, a more structured use of this time could reduce the instructor's workload. For example, during the first measurement at 3 K, the students could begin drafting their reports. They could sketch the experimental setup, review the relevant theory, or begin preparing their data analysis scripts. This would keep them productively engaged without requiring interaction with the TA.

During the 300 mK measurement, students have the opportunity to observe the formation of the first clear Hall plateaus in real time. This is a highlight of the experiment and led to good discussions during my observation. However, once the initial plateaus have appeared, the formation of the rest becomes repetitive. At that point, it would make sense for students to shift their focus to analyzing data from the first measurement. Structuring the workflow in this way would reduce the need for the TA to keep students engaged through conversation, and would potentially allow multiple groups to work in parallel, provided the hardware constraints are addressed. This could be implemented in Nano 3 as well.

One specific addition I would recommend for future versions of the experiment is including a calculation of the electron mean free path in the student report. Based on the results from my own measurements, the mean free path corresponds to electrons passing roughly 3,000 atoms before scattering, assuming an atomic spacing of 5 Ångström. This gives students an intuitive and memorable sense of the scale involved in nanostructure physics, and emphasizes the remarkable cleanliness of the Hall bar sample used in the experiment.

On the other hand, I do not believe it would be practical to add a measurement of the effective mass. Although relevant from a theoretical perspective, it is very time-consuming in practice. It took me four hours to perform five magnetic field sweeps, since I returned to the same starting point each time to be able to compare the different measurements. While it might be possible to speed up the process by alternating magnetic field sweep directions, the overall time investment would still be substantial, and the measurement would require its own lab session. If this measurement was to be included in future iterations of Nano3, I would recommend removing one of the preliminary experiments.

9 Conclusion

The purpose of this project was to investigate how the newly established Quantum Training Lab (QTL) at the Niels Bohr Institute can be used to teach students about quantum phenomena through hands-on experimentation. The focus was on the Quantum Hall Effect (QHE), which served as the final experiment in the course "Quantum Phenomena in Nano Systems (Nano3)".

In order to evaluate the educational value of the experiment, I observed a group of students as they performed it and interviewed them afterwards. The students expressed enthusiasm about working with advanced equipment and appreciated the opportunity to apply their knowledge of quantum mechanics in a practical context. They showed an increased understanding of the QHE, including the difference between classical and quantum regimes, and the significance of low temperature and high magnetic field. Although the experiment required relatively little manual work, the students still felt that they had gained relevant experimental experience regarding their ability to replicate the experiment, and insight into how

one tests physical theories in practice.

To further understand the setup and the underlying physics, I personally performed the experiment and carried out a more detailed analysis than was required of the students. This included measuring the effective mass of the electrons in the sample using temperature-dependent Shubnikov-de Haas oscillations, and calculating the mean free path and scattering time. The extracted value of the effective mass was consistent with expectations.

Based on my experience with the setup and observation and interview of the students, I proposed ideas for how the experiment could be implemented in larger courses, as well as changes that could be implemented in next year's Nano3 course. Regarding the implementation in larger courses, I recommended restructuring the workflow to make better use of down-time in between and during measurements. I also recommended highlighting the formation of the plateaus seen in the 300 *mK* measurement to keep students engaged, and letting them see a quantum phenomenon in real time. For next year's Nano3 course I recommended delving deeper into the data analysis, but without introducing additional measurements. While the current format works well for a small course like Nano3, certain logistical constraints will need to be addressed, if the experiment is to be scaled for larger courses.

Overall, the experiment performed at the QTL provided a valuable opportunity for students to engage with quantum mechanics in a concrete and meaningful way. The combination of theoretical understanding, experimental technique, and modern equipment makes the QHE experiment a strong candidate for being part of the training material for the upcoming quantum workforce. Once the QTL is completely set up, I expect that a significant part of future physics students pursuing a degree in quantum physics will see quantum phenomena with their own eyes inside the laboratory.

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Appendix A

Table A.1: List of variables used in the project

Variable	Description	Unit
e	Elementary charge	C
ρ	Resistivity	Ω
σ	Conductivity	S
h	Planck constant	J · s
$\hbar$	Reduced Planck constant	J · s
m^*	Effective mass	-
n	Electron density	m^{-2}
μ	Mobility	cm^2/V
τ	Scattering time	s
l_e	Mean free path	m
k_B	Boltzmann constant	J/K
ω_B	Cyclotron frequency	Hz
l_B	Magnetic length	m

Appendix B

When a two-dimensional system such as a 2DEG is subject to a strong perpendicular magnetic field, one where $\omega_B \tau \gg 1$, the quantization of states leads to an oscillatory resistivity [3]. The effect is known as the Shubnikov-de Haas effect and the oscillations therefore bear the name Shubnikov-de Haas oscillations. These are primarily evident in the longitudinal resistivity ρ_{xx} and can be seen in fig. 7.1. The longitudinal resistivity oscillates around the classical resistivity, and the oscillations are periodic in $1/B$ [3].

Appendix C

For a system of size $L_x \times L_y$, the degeneracy of each Landau level can be calculated. In the x -direction there are plane waves, so it is equivalent to the particle in a box problem. Here the x -momentum is quantized with the distance $\Delta k_x = \frac{2\pi}{L_x}$ between each oscillator. In the y -direction the oscillators are localized around $y_0 = k_x l_B^2$. The center of the oscillator should lie inside the sample so a logical constraint would be $0 \leq y_0 \leq L_y$. Since y_0 is directly linked with $\hbar k_x$ this imposes a constraint on allowed values for k_x : $0 \leq k_x \leq L_y/l_B^2$ [7]. The total number of oscillators per Landau level can now be calculated as

$$N = \frac{L_x}{2\pi} \int_0^{L_y/l_B^2} dk = \frac{L_x L_y}{2\pi l_B^2} = \frac{A|e|B}{2\pi\hbar} = \frac{AB}{h/|e|}. \quad (C.1)$$

Appendix D

As part of the course the Nano3 students were required to submit a report on the QHE experiment. As part of the report they had to analyze their data in order to answer some questions. The following is an excerpt of these questions:

- Calculate the carrier density n and Hall mobility μ from the data in the classical regime.

- Discuss the differences between the data taken at the two temperatures. In a separate paragraph, provide your interpretation.
- Calculate and make a graph of the longitudinal resistivity ρ_{xx} and Hall resistivity ρ_{xy} versus the magnetic field B . The resistance quantum $R_K = h/e^2 = 25.8 \text{ k}\Omega$. Discuss the multiples of R_K at which the plateaus in ρ_{xy} lie.
- How many plateaus can you distinguish in ρ_{xy} ? What is the spacing between them? Are they periodic in B ? What about $1/B$? Make a graph of the plateau index vs. B and $1/B$.
- In the previous labs, you extracted the Planck constant using a lightbulb and a few LEDs. Now, what values do you get from the measured ρ_{xy} data?

Appendix E

Propagation of uncertainties

For a function of n independent variables $f(x_1, x_2, \dots, x_n)$ the errors of each variable can be combined using the law of combinations of errors, also known as propagation of uncertainties, to find σ_f which is the uncertainty of f [17]

$$\sigma_f = \sqrt{\left(\frac{\partial f}{\partial x_1}\right)^2 \sigma_{x_1}^2 + \left(\frac{\partial f}{\partial x_2}\right)^2 \sigma_{x_2}^2 + \dots + \left(\frac{\partial f}{\partial x_n}\right)^2 \sigma_{x_n}^2}. \quad (\text{E.1})$$

Weighted mean

For measurements x_i of the same quantity with known uncertainty σ_i the weighted mean can be calculated by [17]

$$\bar{x} = \frac{\sum_{i=1}^n \left(\frac{x_i}{\sigma_i^2}\right)}{\sum_{i=1}^n \frac{1}{\sigma_i^2}}, \quad (\text{E.2})$$

and the standard error of the weighted mean by [17]

$$\sigma_{\bar{x}} = \sqrt{\frac{1}{\sum_{i=1}^n \frac{1}{\sigma_i^2}}}. \quad (\text{E.3})$$

Varying ramp speed when measuring SdH oscillations

In fig. 7.1 plot b and d have fewer points than a, c, and e, because I used a higher ramp speed for b and d. For a, c, and e I used a ramp speed of 0.1 T/min , while for b I used 0.2 T/min , and for d 0.15 T/min .

This was not intended! After measurement a I had some problems with the program that ran the measurement and I ended up rebooting it. I clicked run all to start up the program, but I had forgotten that the default ramping speed of the program was higher than the one I was using for this experiment (which was requested by the lab lead). After the second measurement b I changed the ramping speed back to the one used in a and ran the measurement again. At the time, I had not considered that an increased ramping speed would result in fewer data points, and for the fourth measurement d I increased the ramping speed a bit, despite the lab leads advice, since I was running out of time. During this measurement I realized that

this meant fewer data points and so for the fifth measurement e I went back to the original ramping speed of 0.1 T/min .

Additional plots

In fig. 7.1 plot f shows the peaks of the SdH oscillations and the sine curves fitted to them. For each peak α , β , and γ the data points measured at different temperatures are plotted overlaying each other. In fig. E.1 the measurements of each peak at each temperature are plotted together with the corresponding fitted sine curve.

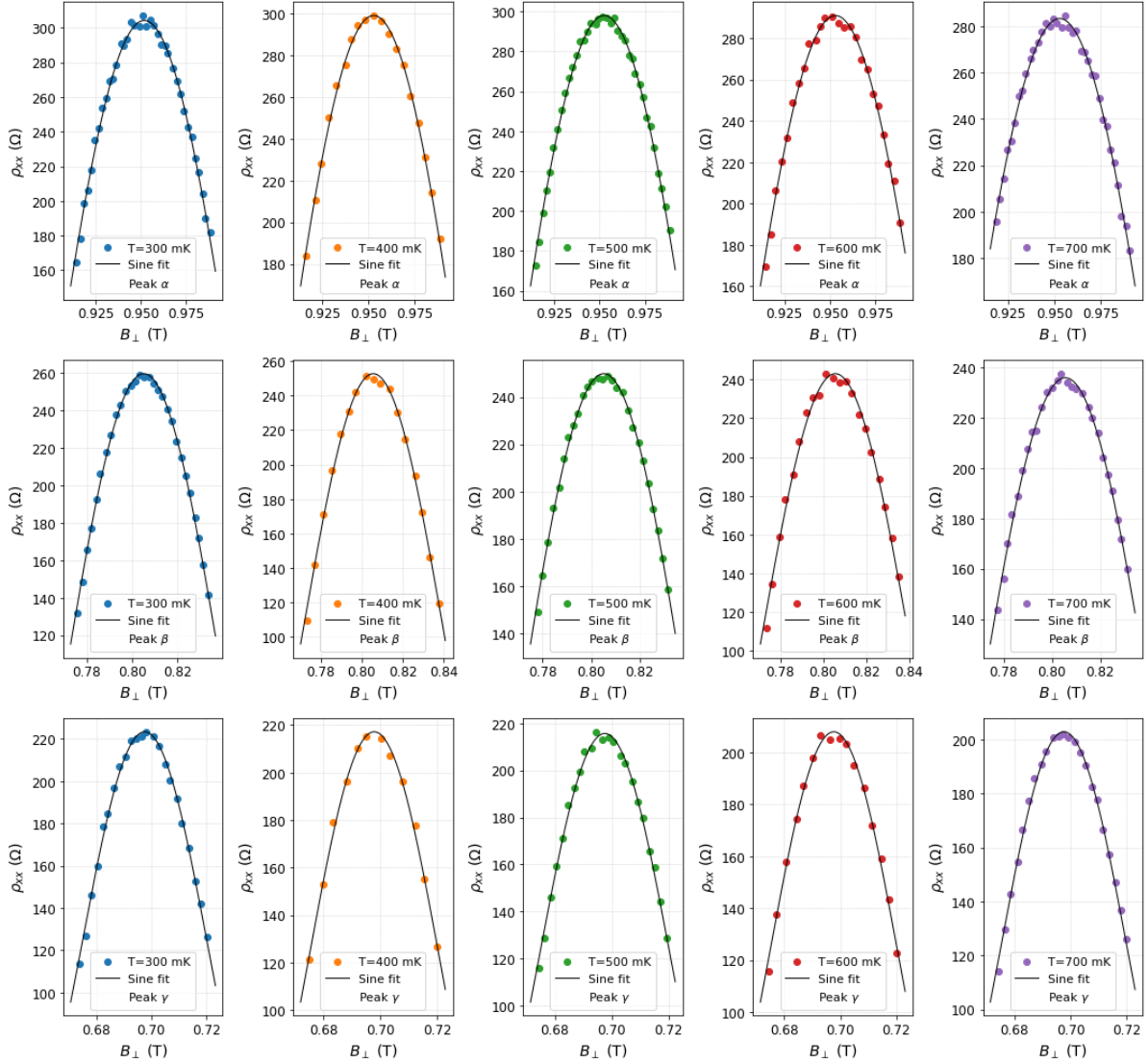


Figure E.1: Measurements of each peak at each temperature plotted together with the corresponding fitted sine curve